Media Engineering and Technology Faculty
German University in Cairo





Sentiment-Enhanced Deep Stock Market Forecasting Using Bilingual Financial News: An Egyptian Market Case Study

Bachelor Thesis

Author: Hussien Haitham Hussien Abdelmotaleb Hussien
Supervisors: Dr. Mervat AbuElkheir
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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowlegement has been made in the text to all other material used

Hussien Haitham Hussien Abdelmotaleb Hussien
31 March, 2026

Acknowledgments

Acknowledgment goes here.

Abstract

This thesis investigates whether bilingual financial-news sentiment can support short-horizon stock direction forecasting for selected Egyptian Exchange equities. The study focuses on Commercial International Bank, Talaat Moustafa Group, and Orascom Investment Holding, using Arabic and English company-linked news from Mubasher. The work was motivated by an industrial research question from Robin, the data-science arm of Beltone Holding: whether sentiment signals can add predictive value beyond historical market data while remaining interpretable and reproducible.

The implemented pipeline collects and cleans bilingual news, routes English articles to FinBERT and Arabic articles to an AraBERT-based sentiment route, weights article sentiment by relevance and confidence, aggregates article-level predictions into lagged daily features, and evaluates all-news, event-only, and event-weighted representations under chronological walk-forward validation. The final sentiment-only benchmark uses logistic regression for direction prediction and ridge regression for future-return prediction across 1, 3, and 5 trading-day horizons. Leakage, coverage, and overfitting checks are treated as part of the experimental design rather than as optional post-processing.

The main result is intentionally conservative. The strongest accepted sentiment-only candidate is OIH at the 5-day horizon under the all-news bilingual setting, with test accuracy 0.532, balanced accuracy 0.503, F1 score 0.630, and return sign accuracy 0.585. Event-aware variants produced isolated gains but did not improve performance consistently. A separate historical-model integration experiment showed that adding sentiment to a price-based EGX forecasting model improved raw accuracy in 5 of 9 ticker-model comparisons, but the gains were conditional rather than uniform. A sentiment-return correlation study found no stable monotonic association under the pre-specified threshold. An Arabic financial AraBERT adaptation experiment reduced held-out masked-language-model loss by 12.01% relative to base AraBERT, supporting domain-adaptation feasibility but not supervised sentiment superiority. Finally, a social-media feasibility study concluded that Reddit and X/Twitter should remain future work unless official approved access is available.

Overall, the thesis finds that bilingual financial-news sentiment contains weak and unstable information for EGX forecasting in the current dataset. Its strongest contribution is therefore methodological: a reproducible, language-aware, leakage-safe framework for testing sentiment signals honestly in a sparse emerging-market setting.

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Chapter 1

Introduction

Stock markets are not only trading venues. They are mechanisms for capital allocation, corporate financing, and risk transfer. When public information is absorbed into prices, the speed and quality of that absorption affect investors, listed firms, and the broader economy. For research desks and brokerage institutions, the practical question is therefore not merely whether prices move, but whether new information can be interpreted quickly enough to support better decisions before the reaction is fully reflected in the market. This economic relevance is one reason why stock forecasting remains a persistent research problem. [1, 2, 3]

The technical difficulty begins with an information asymmetry. Numerical market series record the market reaction after it occurs, whereas news, disclosures, and commentary describe the events that may have triggered that reaction. A price chart reveals that a move happened. It does not explain whether the move followed earnings news, financing plans, governance changes, or contract announcements. This gap motivates the use of text as an exogenous signal rather than relying on price history alone. [4, 5, 3]

Existing research has therefore moved from price-only forecasting toward multimodal forecasting that combines market variables with textual sentiment, event representations, or contextual language-model features. The evidence is encouraging but not uniform. Some studies report measurable gains from financial news or social-media sentiment, while others show that those gains depend strongly on the text source, the forecast horizon, the market under study, and the evaluation protocol. The relevant scientific question is conditional: under what data and validation conditions does sentiment add predictive value? [6, 7, 8, 9]

This question is especially important in the Egyptian market, where firm-related financial coverage appears in both Arabic and English. A forecasting pipeline that ignores one of the two languages discards part of the information environment. A pipeline that combines both languages without careful routing and relevance control can instead amplify noise. The present thesis emerged from an industrial bachelor collaboration with Robin, the data-science arm of Beltone Holding, where this issue was practical rather than abstract. The working need was to test whether bilingual financial-news sentiment could support EGX-oriented forecasting analysis under a setting that remained interpretable, conservative, and reproducible. Prior work on finance-specific English sentiment modeling, Arabic-specialized language models, and conservative sentiment-evaluation design supports that emphasis on language-aware modeling and strict temporal alignment. [10, 11, 12, 9]

1.1 Problem Statement

This thesis investigates whether sentiment extracted from bilingual financial news can improve short-horizon stock direction forecasting in order to determine the predictive value of sentiment-derived signals for selected Egyptian Exchange equities, namely CIB, TMG, and OIH.

1.2 Motivation

The industrial context of the study is a strong motivation for the chosen design. In the Robin and Beltone workflow that motivated the thesis, analysts face a mixed Arabic-English news stream, uneven company coverage, and many articles that are attached to a company page without being equally relevant to valuation. A useful forecasting signal in that setting must therefore do more than assign positive or negative labels. It must route each language to an appropriate sentiment model, distinguish direct company coverage from weaker mentions, and aggregate articles in a way that preserves timing and relevance. This strategy is also consistent with prior work showing that source choice, finance-specific English sentiment modeling, and Arabic-specialized encoders materially affect the quality of textual sentiment signals. [5, 10, 11]

The second motivation is methodological. A strong price-based forecasting system can hide whether sentiment is truly informative or whether it is merely a small additive feature inside an already expressive model. For that reason, the thesis adopts a sentiment-only benchmark before any later integration with richer market features. This choice makes the interpretation cleaner and aligns with the cautionary literature on sentiment forecasting, where apparent gains often disappear once feature leakage, horizon choice, or loose validation procedures are controlled more strictly. [9, 8, 3]

1.3 Research Questions

RQ1. To what extent does bilingual financial-news sentiment contain predictive information for short-horizon EGX stock direction forecasting?

RQ2. How do language-specific routing, article relevance, and event-aware aggregation affect the quality of the extracted sentiment signal?

RQ3. Do any observed predictive gains remain valid under strictly chronological and leakage-safe evaluation?

RQ4. What do auxiliary experiments with historical-model integration, sentiment-return correlation, Arabic financial-domain adaptation, and social-media feasibility reveal about the future development path of EGX sentiment forecasting?

1.4 Objectives

Objective 1. Design a bilingual financial-news forecasting formulation for selected EGX equities.

Objective 2. Develop a sentiment extraction framework that distinguishes Arabic and English article processing.

Objective 3. Construct sentiment features that account for article relevance, language composition, and event structure.

Objective 4. Evaluate the resulting forecasting signal under chronological validation and test splits with explicit leakage controls.

Objective 5. Investigate whether the observed sentiment signal is strong enough to justify further sentiment-enhanced forecasting work in the Egyptian market.

Objective 6. Document auxiliary experiments that test sentiment as an added feature in a historical model, inspect direct sentiment-return association, evaluate Arabic financial-domain adaptation, and define ethical future work for social-media sources.

1.5 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 establishes the background concepts required for the study, including forecasting targets, class imbalance in directional prediction, financial text as an exogenous information source, transformer-based sentiment extraction, and leakage-safe evaluation. Chapter 3 then reviews state-of-the-art work on sentiment-based stock forecasting and positions the present thesis within that literature. Chapter 4 presents the solution approach adopted in the study. Chapter 5 reports the implementation choices, experimental setup, and empirical findings. Finally, Chapter 6 summarizes the main conclusions and outlines future research directions.

Chapter 2

Background

2.1 Financial Forecasting as a Problem of Limited Predictability

Stock forecasting sits at the intersection of economics, statistics, and machine learning. Its difficulty begins with the nature of the market itself. Financial prices respond to heterogeneous information, strategic trading, liquidity effects, and macroeconomic shocks. The efficient-market literature formalized this challenge by arguing that public information is rapidly incorporated into prices. Later empirical work showed that the random-walk view is too strict to hold universally, but it also confirmed that predictable structure in market data is weak, unstable, and difficult to exploit consistently. [1, 2]

This tension explains why forecasting remains an active research area. If markets were perfectly predictable, the problem would already be solved. If they were perfectly unpredictable, forecasting models would be pointless. In practice, the research problem lies between these extremes. Small predictive patterns may exist, but they are noisy, regime-dependent, and sensitive to how the prediction target is defined. Surveys of the field show that forecasting studies differ substantially in horizon, label construction, feature design, and evaluation protocol. Those choices often matter as much as the choice of learning algorithm itself. [13, 3]

2.2 Forecast Targets and Input Modalities

Financial forecasting studies use several related targets. Some predict the next price level. Others predict return magnitude, return sign, or discrete upward versus downward direction. Directional forecasting is attractive because it aligns naturally with decision-making, but it also discards part of the information contained in the return magnitude. Regression targets preserve more information, yet they are often harder to estimate accurately. For this reason, many studies evaluate both direction and return as complementary views of the same forecasting task. [3, 14, 15]

The corresponding input spaces also vary. Traditional models rely on historical prices, returns, volume, and hand-crafted technical indicators. These variables summarize past

market behavior, but they do not describe the semantic content of new information. A news article about profits, financing, or governance may precede a price move even when the numerical history alone is ambiguous. This observation motivates the use of exogenous text signals in addition to market data. The broader forecasting literature increasingly treats stock prediction as a multimodal problem rather than a purely numerical one. [13, 3, 14]

2.3 Class Imbalance and Evaluation Sensitivity

Directional forecasting also creates a statistical issue that is easy to underestimate: class imbalance. Upward and downward moves are rarely distributed evenly across a financial sample. Persistent market drift, low-volatility periods, and sparse firm-level event coverage can all push the labels toward one class. In that setting, raw accuracy can overstate model quality because a classifier can appear successful while still failing on the minority class. For that reason, this thesis reports balanced accuracy, F1, and confusion-matrix statistics alongside headline accuracy, in line with the broader evaluation caution emphasized in recent forecasting studies. [3, 9, 16]

The problem becomes sharper in sentiment-based firm-level studies. News coverage is uneven across firms, and longer forecast horizons can create clusters of same-direction labels. A model may then learn the dominant class more easily than it learns a robust relation between sentiment and future movement. Table 2.1 summarizes why imbalance matters and which safeguards are needed before a result can be interpreted as economically meaningful.

2.4 Textual Information and Sentiment in Finance

The finance literature has long treated text as an information-bearing object rather than as unstructured noise. Early work examined whether media tone, message-board language, or corporate disclosures could explain subsequent market behavior. Negative media language was shown to correlate with market activity and price pressure. Message-board sentiment was also found to contain information about volume and volatility, although the signal quality depended on aggregation and filtering. Finance-specific dictionaries later showed that general-purpose sentiment lexicons can misclassify domain vocabulary, which is why terms such as “liability” or “capital” require financial context to be interpreted correctly. [4, 6, 17, 5]

These findings motivate a distinction between sentiment source and sentiment quality. Not every text stream has the same predictive value. Financial news is usually more structured than social-media text and is more likely to contain firm-linked events. Social media is timelier and larger in volume, but it is also noisier and harder to map to a specific asset. Survey work in financial sentiment analysis therefore emphasizes that source choice, content analysis method, and aggregation strategy jointly determine whether textual sentiment becomes a useful explanatory variable. [5, 18, 7]

Table 2.1: Why class imbalance is a methodological concern in directional financial forecasting.

Imbalance source	Financial forecasting manifestation	Risk if ignored	Appropriate safeguard
Market drift	One direction dominates over a sample period, especially in short binary up/down labels	Raw accuracy overstates true discrimination	Balanced accuracy and confusion-matrix reporting
Sparse firm coverage	Company panels with few relevant articles generate unstable minority-class observations	High F1 can be driven by majority-class prediction	Per-ticker evaluation and coverage analysis
Longer target horizons	Multi-day returns can cluster on one side during sustained moves	Thresholds become biased toward the dominant class	Validation-only threshold tuning and horizon-specific reporting
Event concentration	Market-moving news arrives in bursts rather than uniformly across dates	A model can memorize event periods instead of learning a stable relation	Chronological splits with embargo and walk-forward validation

2.5 From Lexicons to Deep Financial NLP

Text-based market prediction has evolved from sparse lexical features toward distributed and contextual representations. Early financial news systems used bag-of-words, noun phrases, named entities, and support-vector models. Schumaker and Chen's AZFinText system used breaking financial news and intra-day stock data to predict short-term price movement, showing that structured textual features can support directional prediction. Hagenau, Liebmann, and Neumann later introduced context-capturing features and reported improvements over simpler textual baselines for stock-price prediction from financial news. These systems demonstrated that the representation of text strongly influences the usefulness of the downstream forecasting model. [19, 20]

Deep learning extended this line of work by learning latent semantic features directly from text. Event-driven approaches represented news events in dense form and modeled how short-term and long-term event effects influence price movement. Other models fused news with recurrent architectures, attention mechanisms, or hybrid text-price representations. These studies moved the field from manually designed textual cues toward learned semantic encodings. They also made clear that the meaning of a financial article cannot be reduced to isolated polarity words alone. [21, 22, 23, 24]

The broader NLP shift toward transformer-based models is also central to this thesis. The transformer architecture replaced recurrent sequence processing with self-attention, enabling stronger contextual representations at scale. BERT then showed how pre-trained bidirectional language models could be adapted to downstream tasks with minimal archi-

tectural change. Domain-specialized variants followed, including FinBERT for English financial text. This progression matters because sentiment extraction quality increasingly depends on pre-training domain and language rather than on task-specific feature engineering alone. [25, 26, 10]

2.6 Arabic and Bilingual Financial Text Processing

Arabic financial NLP introduces additional complexity. Arabic is morphologically rich, orthographically variable, and distributed across Modern Standard Arabic, dialectal Arabic, and specialized domain language. These properties make direct transfer from English sentiment models unreliable. Arabic-specific pre-training has therefore become important for downstream classification quality. AraBERT demonstrated strong gains over multilingual baselines on Arabic NLP tasks, while ARBERT and MARBERT showed that specialized Arabic pre-training can outperform larger multilingual models on a broad benchmark of Arabic understanding tasks. [11, 12]

Bilingual settings introduce a further modeling decision: whether to force all text through a single multilingual model or to route each language through a more suitable encoder. Recent Arabic model studies indicate that language variant and task proximity can be more important than raw pre-training size. CAMeLBERT provides additional evidence that Arabic model specialization benefits downstream performance when the pre-training data better matches the target task. For a bilingual financial-news pipeline, these findings justify treating language routing as a methodological design choice rather than a preprocessing detail. [27, 12, 11]

2.7 Chronological Evaluation and Leakage

The final background concept is evaluation discipline. Time-series forecasting is unusually vulnerable to leakage because future information can enter the feature space in subtle ways. The leakage may come from same-day article alignment, overlapping target windows, hyperparameter tuning on test periods, or refitting logic that uses observations whose targets would not yet be known at prediction time. Reported performance can therefore be overstated even when the implementation looks reasonable on the surface. [9, 8]

This concern is especially relevant for text-based forecasting. News has timestamps, trading occurs on exchange calendars, and forecast horizons span multiple future observations. A method that ignores those interactions can produce optimistic but invalid results. Chronological train-validation-test splits, embargo regions between adjacent blocks, and walk-forward evaluation are therefore not optional refinements. They are part of the problem definition itself. This thesis adopts that view and treats leakage control as a core experimental requirement rather than as a final diagnostic step. [9, 3]

2.8 Chapter Summary

This chapter established the conceptual foundations for the thesis. Stock forecasting is difficult because market predictability is limited and unstable. Textual information is

attractive because it describes events that numerical histories do not represent directly. The usefulness of that information depends on representation quality, label balance, language suitability, and evaluation discipline. These background concepts motivate the more specific state-of-the-art review in the next chapter, which examines how prior work operationalized these ideas and where the present thesis contributes.

Chapter 3

Related Work

3.1 From Price-Only to Text-Aware Forecasting

Research on stock prediction did not begin with sentiment. Early systems relied on prices, returns, volume, and technical indicators because these variables are directly observable and easy to align temporally. The main weakness of that tradition is explanatory rather than computational. Numerical series encode market state, but they do not encode the semantic content of the events that moved the market. Once researchers began to treat news and social media as external information channels, forecasting became a multimodal problem rather than a purely numerical one. [14, 15, 3]

The shift from price-only to text-aware prediction followed a recognizable sequence. Schumaker and Chen's AZFinText system used breaking financial news to support short-term directional prediction and showed that textual features can add value when the entity linkage is explicit and the horizon is short. Hagenau, Liebmann, and Neumann then improved on simple bag-of-words pipelines by introducing context-capturing features for financial news classification. Later work replaced manual text features with event embeddings, hierarchical attention, and latent-variable sequence models. Ding et al. modeled structured events, Hu et al. modeled news sequences with attention, and Xu and Cohen combined tweets with price history in a deep generative framework that reported state-of-the-art performance on a newly collected dataset. The common lesson is that text representation, entity resolution, and temporal alignment are as important as the downstream predictor. [19, 20, 21, 22, 24]

3.2 Social Media Sentiment

Social media has attracted sustained attention because it is timely, large in volume, and rich in investor reaction. Bollen, Mao, and Zeng analyzed approximately 9.8 million tweets from February to December 2008 and linked mood dimensions to subsequent movements in the Dow Jones Industrial Average. Their best reported directional result reached 87.6% accuracy after including mood information, but the study was conducted at market-index level rather than firm level. That design is important because aggregate mood can correlate with broad market movement without solving the harder task of mapping noisy user text to specific equities. [7]

Nguyen, Shirai, and Velcin moved closer to firm-level prediction by using company-related message-board data and topic-aware sentiment features. Their experiment covered 18 stocks over one year, with 171 transaction dates for training and 78 for testing. Relative to a price-only baseline, their topic-sentiment method improved mean accuracy by 2.07 percentage points across the 18 stocks and by 9.83 percentage points on the subset of difficult stocks. This is a useful result because it quantifies the value of better ticker-specific filtering. Its limitation is that the setup remains short-horizon, single-language, and dependent on shallow topic extraction rather than contextual language models. [18]

Mao, Counts, and Bollen broadened the comparison by evaluating survey sentiment, news headlines, Twitter data, and search-engine signals in the same predictive study. Their main result was not that every online signal helps, but that different sources behave differently in time. Search and social signals were more informative for short-run market behavior than traditional survey measures. This distinction matters because it suggests that sentiment source is not an interchangeable design choice. Speed, topic granularity, and user noise all affect what a sentiment variable actually measures. [28]

Taken together, the social-media literature shows that sentiment can carry predictive information, but only after heavy filtering. Short text length, sarcasm, reposted content, unofficial company references, and bot-driven amplification all weaken the link between message sentiment and tradable firm-specific information. These constraints are especially relevant for emerging markets with bilingual discussion, because entity linking and language variation become harder before modeling even begins. [7, 18, 29]

Arabic social-media forecasting work is especially relevant to the future direction of this thesis. AlKhatib et al. studied Arabic tweets and stock-market changes in the Arab world, including EGX30, using correlation and Granger-causality style analysis. Their study supports the idea that Arabic investor discussion may contain market-relevant information for Egyptian equities. However, it also reinforces the need for careful source access and entity filtering: social-media relevance depends on whether the collected posts can be legally obtained, temporally aligned, and linked to the correct market or company. [30]

3.3 Financial News Sentiment

Financial news provides a cleaner signal source than social media because the language is more structured and the company linkage is more explicit. Mohan et al. assembled a five-year dataset of daily prices for S&P 500 firms together with more than 265,000 related financial news articles, then compared deep models built from price data, sentiment information, and hybrid text-price inputs. The scale of the dataset is a strength because many earlier sentiment studies were limited by sparse article counts. The main limitation is external validity: a large U.S. large-cap news panel is a much easier information environment than a thinner bilingual EGX news stream. [23]

Dahal et al. addressed a different problem. Rather than mixing many architectures and feature sets, they compared LSTM and GRU models under identical conditions with and without financial news sentiment. Performance was assessed with RMSE, MAPE, and the correlation coefficient R. Their conclusion was that the sentiment-augmented variants outperformed stock-only variants, which is methodologically useful because it

isolates the effect of adding text. At the same time, the study focuses on regression quality for developed-market data, not on leakage-sensitive directional classification for sparse firm-level coverage. [8]

Ho and Huang fused sentiment analysis with candlestick-chart representations in a two-branch convolutional architecture. Their experiments covered five high-demand stocks, namely Apple, Tesla, IBM, Amazon, and Google, and they evaluated forecast windows of 4, 6, 8, and 10 days. The best reported accuracy was 75.38% for Apple. This study is useful because it demonstrates that representation fusion can outperform either visual price input or sentiment input alone. Its limitation is equally clear: the result is stock-specific, U.S.-centric, and based on social-media sentiment rather than a firm-linked financial-news archive. [31]

Cristescu et al. used market-news sentiment as an exogenous factor in regression models for stock opening prices. Their most relevant contribution is not a headline accuracy number but the explicit goodness-of-fit analysis. One reported NARX specification achieved an ANOVA significance level of $p = 0.020$, which supports the claim that sentiment can improve model fit in that setting. The limitation is that regression fit is not the same as robust directional forecasting. A model can improve opening-price estimation while still failing to discriminate reliably between upward and downward moves. [32]

Zhao and Yang proposed a deeper fusion system, SA-DLSTM, that combines an emotion-enhanced CNN, a denoising autoencoder, and an LSTM for stock movement prediction on the Hang Seng Index. The paper is important because it captures several design ideas that recur in later literature: sentiment-strength weighting, feature denoising, and sequence modeling over multivariate financial inputs. Its main limitation is that the target is still an index-level market series. That setting avoids the more difficult problem of article-to-firm relevance matching, which is central in the present thesis. [33]

Liapis et al. provide the most useful benchmark-oriented reference for this thesis. Their study evaluates 30 forecasting methods over 67 feature setups and three forecast horizons. The strongest conclusion is not that one sentiment method wins, but that no single sentiment configuration dominates all horizons. This is an important correction to the literature because it shows that sentiment effectiveness is conditional on representation and horizon choice. It also reinforces the need for conservative reporting. A sentiment setup that performs well for one horizon may be statistically unremarkable or even weak at another. [9]

3.4 Why Sentiment Does Not Always Improve Forecasting

The literature does not support a simple claim that sentiment always improves predictive accuracy. Several failure modes recur. The first is semantic coarseness. A positive article can describe a low-impact corporate event, while a seemingly neutral article can contain material financing or governance information. Mudinas, Zhang, and Levene argue that sentiment attitudes alone do not consistently explain future price changes and that any useful relation is strongly case-specific. This is a direct warning against treating polarity as a sufficient event representation. [34]

Table 3.1: Quantitative comparison of representative sentiment-based stock forecasting studies.

Study	Market and sample	Text input	Model family	Reported result	Main limitation
Bollen et al. [7]	DJIA, about 9.8 million tweets, Feb.–Nov. 2008	Twitter mood streams	GPOMS mood features with SOFNN	87.6% direction accuracy	Index-level setup with no firm-level entity mapping
Nguyen et al. [18]	18 stocks, 1 year, 171 train dates and 78 test dates	Message-board topic sentiment	Topic-sentiment features with linear SVM	+2.07 percentage points mean accuracy over price-only; +9.83 on difficult stocks	Short horizon, shallow topic model, single-language data
Mohan et al. [23]	S&P 500 firms, 5 years, more than 265,000 news articles	Financial news	Deep text, price, and fusion models	Large-scale comparison across price-only, text-only, and fusion settings	Developed-market large-cap panel with no bilingual routing problem
Dahal et al. [8]	Developed-market stock data with matched financial news	Financial news plus stock features	LSTM and GRU	Sentiment-augmented models improved RMSE, MAPE, and R over stock-only variants	Regression emphasis, no firm-level bilingual challenge
Ho and Huang [31]	Five U.S. stocks, 4 to 10 day horizons	Social-media sentiment plus candlestick charts	Two-branch CNN	Best reported accuracy 75.38% for Apple	Stock-specific result and social-media input rather than financial-news input
Cristescu et al. [32]	Selected stocks with lagged opening-price models	Market news sentiment	Linear and NARX regression	One reported NARX fit significant at $p = 0.020$	Goodness of fit does not guarantee robust directional discrimination
Zhao and Yang [33]	Hang Seng Index and public opinion streams	Online sentiment plus market variables	ECNN + DAE + LSTM	Higher accuracy than baselines in the integrated framework	Index-level setting avoids article-to-firm relevance matching
Liapis et al. [9]	30 methods, 67 feature setups, 3 horizons	Sentiment time series plus prices	Broad benchmark	No single sentiment setup dominated all horizons	Strong evidence that gains are horizon- and setup-specific

The second failure mode is temporal misalignment. If article sentiment is joined to prices on the wrong day, or if a target window overlaps the feature window, reported performance can be inflated. Liapis et al. show that different sentiment feature setups behave differently across horizons, which implies that loose alignment can easily distort comparisons. In other words, methodological noise can be mistaken for predictive signal. [9]

The third failure mode is data scarcity and imbalance. Sparse firm-level article panels create unstable feature histories, and directional labels often lean toward the majority class. A model can then achieve acceptable raw accuracy by overpredicting the dominant direction. This issue is especially relevant for emerging markets where article volume is much lower than in U.S. large-cap datasets. The combination of sparse text and imbalanced labels makes balanced accuracy and confusion-matrix analysis more informative than headline accuracy alone. [16, 9]

The fourth failure mode is source mismatch. Social media is fast and abundant, but

it is noisy and weakly linked to formal firm events. Financial news is cleaner and more structured, but it is slower and less dense. The contrast between Bollen, Nguyen, and Mao on one side and Mohan, Dahal, and Cristescu on the other shows that sentiment source changes both the signal quality and the methodological difficulty. No source is universally superior. The choice must match the research objective, the market, and the entity-linking assumptions. [7, 18, 28, 23, 8, 32]

3.5 Domain-Specific and Arabic Language Models

The quality of the sentiment signal depends heavily on the language model used to produce it. Generic sentiment tools often misread financial language because words such as "liability", "dilution", or "guidance" carry domain-specific meanings. FinBERT was designed precisely for this issue. The original paper reports a 14 percentage point improvement over the previous state of the art on Financial PhraseBank, which makes it a much stronger candidate for financial English than a generic sentiment classifier. This result matters because weak sentence-level sentiment estimation will propagate directly into weaker daily forecasting features. [10]

The Arabic side requires an additional specialization step. AraBERT reported state-of-the-art performance on most tested Arabic NLP tasks and the original evaluation reports improvements of up to 2.53 F1 points on the downstream benchmarks. ARBERT and MARBERT extended this line further and introduced the ARLUE benchmark, where their best model achieved an aggregate score of 77.40 and outperformed XLM-R Large despite the latter being substantially larger. These results make a strong case for routing Arabic text through Arabic-specialized encoders rather than treating Arabic as a secondary case inside a generic multilingual pipeline. [11, 12]

Variant sensitivity is equally important. CAMeLBERT compared Arabic models across five task types and 12 datasets and concluded that variant proximity between pre-training data and downstream data matters more than raw pre-training size. This finding is directly relevant for bilingual EGX forecasting. The problem is not only to classify sentiment, but to classify it in a market where Arabic and English articles occupy different stylistic and domain positions. A single shared model may be simpler to deploy, but the literature suggests that such simplicity often comes at the cost of linguistic fit. [27]

At the same time, the Arabic-model literature leaves an open gap. These models are evaluated on Arabic NLP benchmarks, not on bilingual financial forecasting benchmarks for the Egyptian market. Their strong benchmark performance justifies their use as sentiment encoders, but it does not by itself prove that they will improve stock prediction. That remaining empirical question is one of the contributions of the present thesis. [11, 12, 27]

Recent Arabic financial NLP resources make this gap more visible. SAHM introduces Arabic financial and Shari'ah-compliant reasoning tasks, while recent Saudi-market work studies Arabic financial sentiment with large language models. These papers support the need for Arabic finance-specific evaluation rather than relying only on general Arabic sentiment benchmarks. They also clarify the limitation of the present thesis: without independent human-labeled Arabic financial sentiment examples, AraBERT can be adapted to financial text through masked-language modeling, but it cannot be validly evaluated as a supervised Arabic FinBERT equivalent. [35, 36]

3.6 Evaluation Metrics and Research Gap

Comparison across prior forecasting studies is complicated by inconsistent evaluation practice. Regression-oriented papers often report RMSE, MAE, MAPE, or correlation. Directional papers more often report accuracy, precision, recall, and F1. A few papers add simulated returns, which is useful but depends strongly on trading assumptions. These metrics are not interchangeable. A classifier can show a favorable raw accuracy while failing to recover the minority class. A regression model can reduce RMSE while still predicting the sign of the move poorly. For this reason, metrics must be interpreted in relation to the target definition and the class balance. [8, 9, 16, 33]

The gap addressed by this thesis is therefore both geographical and methodological. Most sentiment-based forecasting studies focus on English text, developed markets, single-language pipelines, or index-level prediction. Very few studies examine bilingual Arabic-English financial-news routing for Egyptian equities, and fewer still evaluate that problem under conservative walk-forward validation with explicit leakage and overfitting checks. The present work narrows the scope to three EGX tickers and a single bilingual financial-news source so that article relevance, language routing, and temporal alignment can be studied under cleaner conditions. The contribution is not to assume that sentiment is universally powerful. The contribution is to test, under a more locally relevant and more tightly controlled design, how much forecasting value bilingual financial-news sentiment actually carries.

3.7 Chapter Summary

This chapter reviewed the state of the art in sentiment-based stock forecasting and identified the main lessons that shape the present thesis. Text can help, but the reported gains depend on source quality, feature design, horizon choice, and validation rigor. Financial-news studies are usually cleaner than social-media studies, yet they remain concentrated in English and in developed markets. Domain-specific financial English models and Arabic-specialized transformers offer strong upstream sentiment encoders, but they have rarely been evaluated together in a bilingual EGX forecasting setting. These gaps motivate the methodological choices developed in the next chapter.

Chapter 4

Methodology

4.1 Research Design

This thesis was carried out as an industrial bachelor project between the German University in Cairo and Robin, the data-science arm of Beltone Holding. The original project brief asked whether sentiment information provides additional predictive value beyond historical market data in financial forecasting. The practical work evolved in two stages.

The first stage extended an existing EGX direction-forecasting system that already used historical prices, engineered technical indicators, and machine-learning models. In exploratory development, adding a sentiment signal appeared to improve directional accuracy by a small absolute margin. That result was useful as motivation, but it did not isolate the value of sentiment itself because the underlying price-based model was already strong and heavily feature-engineered.

The second stage, which forms the core reproducible contribution of this thesis, reframed the problem as a sentiment-only benchmark. The goal was to answer a narrower and more defensible question: if price-derived predictors are removed, how much information can bilingual financial-news sentiment alone provide about future stock direction and return? This design makes the interpretation cleaner. If the sentiment-only benchmark is weak, the study can say so directly. If it is moderately useful, later integration with price-based systems becomes easier to justify.

The final methodology evaluates three complementary ideas. The first is bilingual sentiment extraction from Arabic and English financial news. The second is leakage-safe alignment of article-derived features to later trading dates. The third is the comparison of all-news, event-only, and event-weighted sentiment representations. Figure 4.1 summarizes the end-to-end workflow adopted in the thesis, from bilingual news collection to leakage-safe evaluation.

4.2 Study Universe and Data Sources

The empirical study focuses on three Egyptian Exchange tickers: Commercial International Bank, also written as CIB or COMI; Talaat Moustafa Group, also written as TMG

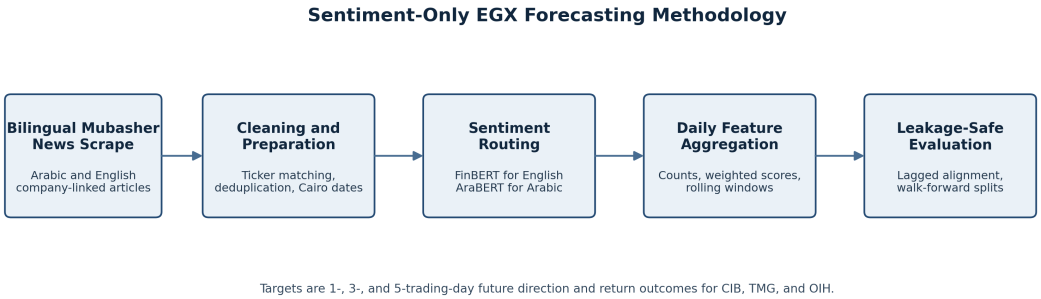


Figure 4.1: End-to-end methodology for the sentiment-only EGX forecasting benchmark.

or TMGH; and Orascom Investment Holding, also written as OIH. These tickers were selected because they are actively discussed in local financial coverage and provide a useful mix of relatively higher-coverage and lower-coverage cases.

The final reproducible benchmark uses Mubasher as the sole news source because it offers both Arabic and English listing pages tied to specific EGX company pages. Restricting the thesis dataset to one source keeps crawling behavior, ticker linkage, and article structure consistent across the study. Historical market prices are loaded from local CSV files stored in the project repository, with external download used only as a fallback if a local file is missing.

The raw scrape and the cleaned article counts differ because the final dataset applies confidence filtering, text cleaning, language normalization, and duplicate removal. Table 4.1 summarizes the final bilingual article coverage used in the thesis.

Table 4.1: Raw scraped records and final sentiment-labeled article counts used in the thesis benchmark.

Ticker	Raw records	Final articles	English	Arabic
CIB	1056	1013	310	703
TMG	609	597	188	409
OIH	697	302	87	215
Total	2362	1912	585	1327

The same distribution is visualized in Figure 4.2. The figure makes the language imbalance visible for all three tickers and shows that OIH has the smallest total article count overall. This asymmetry matters because a sparse text history makes any sentiment-derived predictor less stable and increases the variance of the final estimates.

4.3 News Collection and Article Preparation

The news pipeline begins by scraping Mubasher listing pages separately for Arabic and English. For each ticker-language pair, the crawler collects article identifiers, URLs, titles, summaries, and article bodies. The crawler is incremental and retry-aware so that interrupted collection can be resumed without restarting the archive.

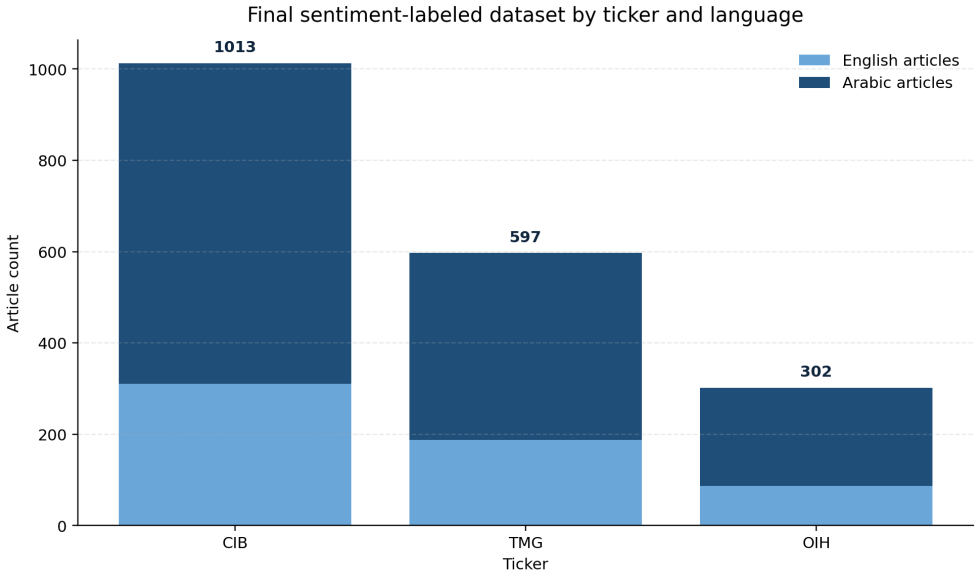


Figure 4.2: Final sentiment-labeled article counts by ticker and language.

Once collected, raw articles pass through a preparation stage that standardizes company references and removes low-quality or duplicated records. In practice, the preparation stage resolves ticker aliases to a canonical identity, normalizes timestamps to Cairo local time, cleans Arabic and English text, removes very short or duplicated records, and filters weak entity matches. The result is a sentiment-ready bilingual article table whose rows correspond to distinct company-linked news items.

Article relevance is then represented explicitly rather than assumed. A direct article contains the company alias in the title. A contextual article contains the alias in the body but not necessarily in the title. An indirect article is linked through weaker alias evidence. A stock-page-only article appears on the company page without stronger textual support. These categories are converted into weights so that strong company references contribute more than weak ones when daily sentiment features are aggregated.

4.4 Bilingual Sentiment Extraction

After article preparation, sentiment is scored using separate models for English and Arabic. English articles are routed to FinBERT [10], while Arabic articles are routed to AraBERT-based sentiment classification [11]. The reason for routing by language instead of forcing both languages through one shared model is methodological. The literature reviewed in Chapter 3 shows that domain-specific financial English models and Arabic-specialized language models are both more defensible than generic multilingual sentiment tools for their respective settings.

For each article, the sentiment layer stores the detected language, the model route, the predicted label, the class probabilities, a confidence estimate, and a continuous sentiment score. The article-level sentiment contribution is then weighted by both relevance and model confidence. This design downweights weak ticker matches and low-confidence predictions before any daily aggregation takes place.

The use of off-the-shelf transformer models is a pragmatic research choice rather than a claim that these models are optimal for Egyptian financial language. The objective of the thesis is to test whether a language-aware bilingual sentiment signal exists at all under a conservative benchmark. If such a signal exists, later work can fine-tune these models on financial-domain Arabic and English corpora.

4.5 Daily Sentiment Feature Construction

Article-level predictions must be converted into time-series features before forecasting can be attempted. The pipeline aggregates scored articles by ticker and Cairo-local calendar date. For each ticker-date pair, it computes article counts, sentiment-label shares, weighted sentiment summaries, confidence summaries, relevance counts, and language-specific summaries in the bilingual view. Short rolling windows and recency indicators are then derived from these daily aggregates so that the model can capture recent sentiment persistence rather than relying on a single daily score.

Three language views are retained throughout the experiments: an English-only view, an Arabic-only view, and a bilingual view that combines both languages after routing and scoring. This separation is important because the Egyptian information environment is genuinely bilingual. The experiments therefore test whether combining both languages adds useful signal or merely adds noise.

4.6 Event-Aware Feature Variants

In addition to standard daily sentiment aggregation, the thesis evaluates event-aware features. The motivation is that articles differ not only in polarity but also in event type, and not all events are equally likely to move a stock. The final pipeline labels articles using broad market-moving categories such as earnings, dividends, mergers and acquisitions, contracts and projects, expansion and investment, financing, and governance or disclosure.

These labels support three feature strategies. In the all-news strategy, every article contributes to the daily sentiment features. In the event-only strategy, only event-related articles are retained. In the event-weighted strategy, all articles remain in the panel but event-related ones receive additional weight. This is not a full thematic NLP model, but it is a practical intermediate step toward richer event-aware relevance modeling.

4.7 Target Construction and Lagged Alignment

The target variables are derived from future closing prices. For each trading day t and forecast horizon h, the future return is defined as

$$r_{t,h} = \frac{P_{t+h}}{P_t} - 1 \tag{4.1}$$

where P_t is the closing price on day t and P_{t+h} is the closing price after h future trading days. A binary direction target is then defined as

$$y_{t,h}^{dir} = \begin{cases} 1, & r_{t,h} > 0 \\ 0, & r_{t,h} \leq 0 \end{cases} \tag{4.2}$$

The thesis evaluates three short forecast horizons. These horizons are reported in the results chapter together with the corresponding implementation settings.

Temporal alignment is one of the most important parts of the methodology. Daily sentiment is shifted before being aligned to a trading date. The features available for trading day t therefore come only from news published before that trading day, not from same-day coverage that could already reflect market movement. This conservative lag is essential for avoiding subtle look-ahead effects.

4.8 Forecasting Framework

The forecasting layer is intentionally simple because the objective is to isolate the informational value of sentiment rather than to maximize performance through model complexity. The thesis therefore uses one low-capacity classifier for direction and one low-capacity regressor for future return. Feature standardization, model selection, and classification-threshold selection are all performed chronologically on validation data only.

Each ticker panel is sorted by date and divided into chronological training, validation, and test regions. Adjacent regions are separated by an embargo that is at least as long as the forecast horizon so that overlapping target windows cannot leak across splits. Final predictions are generated in an expanding-window walk-forward scheme. At every step, the model is trained only on rows whose target values would already be known at the prediction date. This procedure makes the benchmark slower than a random split, but much more credible.

4.9 Evaluation Metrics

The study reports both classification and regression metrics because the task has two outputs: future direction and future return. Directional evaluation uses accuracy, balanced accuracy, precision, recall, F1 score, and confusion-matrix counts. Return evaluation uses mean absolute error, root mean squared error, return sign accuracy, and correlation between predicted and realized returns.

Balanced accuracy is treated as a primary metric rather than an auxiliary one because the direction labels are not expected to be perfectly balanced across all ticker-horizon combinations. A model that leans too strongly toward the majority direction can still appear acceptable under raw accuracy alone. The reported metric set is therefore designed to expose that behavior rather than conceal it.

4.10 Leakage Control and Validity Checks

Because leakage is a common failure mode in financial forecasting, the final pipeline includes explicit validity checks beyond the main experiment loop. These checks verify that the historical rows used for any validation or test prediction do not extend into the future target window of the evaluation row. They also inspect source coverage and compare training, validation, and test behavior so that suspiciously optimistic results are not accepted without further scrutiny.

The overall intention of the methodology is conservative. The thesis is designed to prefer a weaker but believable result over an inflated result produced by loose alignment, optimistic evaluation, or metric choice. That design choice is especially important in an emerging-market setting with limited bilingual data.

4.11 Auxiliary Experiment Protocols

In addition to the final sentiment-only benchmark, the thesis records four auxiliary experiments that explain how the research process evolved and where future work should go. These experiments are not used to replace the accepted final benchmark. Their role is to test surrounding claims and to prevent undocumented exploratory work from being reported informally.

The first auxiliary experiment returns to the original industrial motivation. An external EGX direction-forecasting repository developed during the Robin/Beltone internship was re-run as a historical price-based baseline. The same chronological train-test split was used for each ticker-model pair, and the baseline feature set was compared against an augmented version that added one-day-lagged bilingual sentiment features. This design tests whether sentiment can add value inside a richer market-feature model, but it is treated as exploratory because the target formulation and feature regime differ from the stricter sentiment-only benchmark.

The second auxiliary experiment evaluates direct sentiment-return association. Daily bilingual sentiment is shifted by one calendar day, aligned to later trading dates, and correlated with same-day, 1-day, 3-day, and 5-day returns. Pearson and Spearman correlations are reported together with five-day block-permutation p-values. This experiment is descriptive rather than causal. Its purpose is to check whether the raw sentiment score has a stable monotonic relationship with later returns before more complex forecasting models are interpreted.

The third auxiliary experiment tests Arabic financial-domain adaptation. The Arabic article corpus is used for masked-language-model adaptation of AraBERT, following the first stage of the FinBERT-style domain-adaptation logic. The adapted checkpoint is then evaluated against the original AraBERT model on held-out Arabic financial text using the same masked-token evaluation setup. This is a language-modeling evaluation, not sentiment classification. A supervised Arabic financial sentiment model would require independent human-labeled examples, which are not available in the current thesis dataset.

The fourth auxiliary experiment evaluates social-media feasibility. The target was to find a free, ethical, and officially approved route for adding social-media data. Reddit and

X/Twitter were considered as candidate sources. The protocol rejects unofficial scraping and paid-access bypasses. Social media is therefore kept as future work unless approved official access is obtained and the resulting data can be handled without redistributing raw user content.

4.12 Chapter Summary

This chapter described the solution approach adopted in the thesis. The method begins with bilingual Mubasher news collection, converts cleaned company-linked articles into language-aware sentiment scores, aggregates those scores into daily and event-aware features, aligns them conservatively to later trading dates, and evaluates their forecasting value under chronological walk-forward validation. The chapter also defined the auxiliary experiment protocols used to document the historical-model integration, sentiment-return association, Arabic financial-domain adaptation, and social-media feasibility studies. The next chapter reports the concrete implementation settings and the empirical results obtained from this design.

Chapter 5

Results & Implementation

5.1 Implementation Environment

All final experiments were executed in a Python environment compatible with Python 3.9 or newer. The implementation uses `pandas` and `numpy` for data handling, `scikit-learn` for the forecasting models and evaluation metrics, `requests` and `BeautifulSoup` for the news pipeline, and `torch` together with `transformers` for the sentiment models. The complete project is organized as a reproducible repository with separate modules for the news pipeline, sentiment-only forecasting pipeline, experiment scripts, and validity checks. The public repository is available at https://github.com/hussienhaithamm/EGX-Sentiment-Forecasting-Thesis.

The implementation is driven by saved data and scripts rather than manual notebook edits. Final multi-horizon experiments are executed through a dedicated evaluation script, and leakage, coverage, and overfitting checks are executed through a separate validity script. This separation is useful because it makes the experimental claims auditable at the level of saved outputs rather than relying on informal inspection.

5.2 Final Experimental Dataset

The final sentiment-ready dataset contains 1,912 cleaned and sentiment-labeled Mubasher articles across the three target tickers. Event labeling identifies 1,723 of these articles as belonging to one of the event categories used in the event-aware experiments. Table 5.1 summarizes the ticker-level counts used in the final implementation.

The event counts are high relative to the total article counts, which indicates that Mubasher company pages are strongly event driven rather than dominated by generic commentary. At the same time, coverage remains uneven. CIB has a much denser article history than OIH, and this asymmetry becomes important when the results are interpreted per ticker rather than only in aggregate.

5.3 Experimental Configuration

The final accepted run evaluates three forecast horizons: 1, 3, and 5 trading days. Directional prediction is implemented with logistic regression using balanced class weights,

Table 5.1: Final cleaned dataset and event-labeled subset used in the reported experiments.

Ticker	Final articles	Event-labeled articles
CIB	1013	874
TMG	597	564
OIH	302	285
Total	1912	1723

while return prediction is implemented with ridge regression. Features are standardized before model fitting. The classification regularization parameter is tuned over the grid $\{0.05, 0.1, 0.5\}$, while the regression configuration uses $\alpha = 10.0$. Classification thresholds are not fixed blindly at 0.5. They are selected from validation predictions only and then applied unchanged to the test block.

The chronological setup matches the methodology described in Chapter 4, but the implementation adds the concrete numerical settings required for reproducibility. Each ticker panel is sorted by date, the minimum training history is 250 rows, and walk-forward refits are executed every 250 evaluation rows in the final multi-horizon run. An embargo of at least the forecast horizon is enforced between adjacent data regions so that overlapping target windows do not contaminate validation or test results.

5.4 Main Multi-Horizon Results

Table 5.2 reports the final all-news bilingual results that passed the chronological evaluation pipeline. The table includes the most informative summary metrics for the thesis claims: raw accuracy, balanced accuracy, F1 score, return sign accuracy, and the correlation between predicted and realized future returns.

Table 5.2: Final all-news bilingual sentiment-only results across the three forecast horizons.

Horizon	Ticker	Accuracy	Balanced Acc.	F1	Sign Acc.	Return Corr.
1	CIB	0.491	0.505	0.274	0.474	-0.013
1	TMG	0.520	0.507	0.041	0.469	-0.013
1	OIH	0.461	0.500	0.583	0.426	-0.015
3	CIB	0.514	0.482	0.667	0.516	-0.018
3	TMG	0.511	0.518	0.615	0.493	-0.103
3	OIH	0.471	0.508	0.065	0.518	-0.035
5	CIB	0.543	0.487	0.698	0.510	-0.052
5	TMG	0.493	0.490	0.651	0.532	-0.024
5	OIH	0.532	0.503	0.630	0.585	0.034

The results show a weak and inconsistent sentiment-only signal. The best accepted ticker-horizon pair is OIH at the 5-day horizon, with test accuracy 0.532, balanced accuracy 0.503, F1 score 0.630, sign accuracy 0.585, and return correlation 0.034. The best balanced accuracy in the final accepted run appears for TMG at the 3-day horizon, where balanced accuracy reaches 0.518. These values are above random only marginally. They do not support a claim that bilingual news sentiment alone is a strong standalone EGX forecasting model.

Another pattern is that F1 can look stronger than balanced accuracy. This occurs when the model predicts the majority upward class more often than the minority downward class. For example, several rows show F1 above 0.60 while balanced accuracy remains close to 0.50. This is precisely why balanced accuracy is more informative than raw accuracy or F1 alone in an imbalanced directional setting.

5.5 Event Filtering and Event Weighting

The thesis also evaluates whether event filtering improves the signal. On average, it does not. Relative to the all-news bilingual baseline, the mean event-only accuracy change is -0.013, while the mean event-weighted accuracy change is -0.012. Table 5.3 summarizes these aggregate comparisons.

Table 5.3: Aggregate event-filtering effects relative to the all-news bilingual baseline.

Strategy	Mean accuracy delta vs. all-news	Best positive delta
Event-only	-0.013	+0.025
Event-weighted	-0.012	+0.039

The event-aware variants still produce a few locally interesting cases. CIB improves from 0.491 to 0.507 accuracy at the 1-day horizon under the event-only setting, and from 0.514 to 0.525 at the 3-day horizon. OIH at the 5-day horizon improves from 0.532 to 0.571 accuracy and from 0.630 to 0.706 F1 under event weighting. Yet these improvements are not stable across the full experiment grid. The broader conclusion is that event information can help in selected panels, but it does not produce a consistent accuracy improvement in the final benchmark.

5.6 Validity Checks and Accepted Final Model

The strongest raw-accuracy candidate observed in the final multi-horizon event run was OIH at the 5-day horizon under the event-weighted bilingual setting. Its test accuracy reached 0.571 and its F1 score reached 0.706. However, the corresponding row was flagged by the Phase 7 validity analysis because training accuracy rose to 0.609 while validation accuracy fell to 0.485. The result is therefore exploratory rather than accepted.

The accepted final candidate is OIH at the 5-day horizon under the all-news bilingual setting. Its validation accuracy, balanced accuracy, and F1 score are 0.522, 0.525, and 0.546, respectively. Its test accuracy, balanced accuracy, F1 score, and sign accuracy are 0.532, 0.503, 0.630, and 0.585. This candidate was retained because it passed the leakage and overfitting filters while keeping all three validation and test classification metrics at or above the minimum acceptance threshold.

The Phase 7 validity results are important because they quantify the reliability of the full pipeline. Leakage checks passed for 45 panel evaluations covering 43,880 validation and test prediction rows, and no history violations were detected. Coverage analysis showed that TMG and OIH matched the currently available Mubasher listing pages,

while CIB had one additional Arabic listing page available at audit time. Overfitting analysis flagged 15 of the 45 experiment rows, which confirms that raw test accuracy alone would have overstated the quality of the strongest-looking candidate.

Coverage remains a substantive limitation. Prior-7-day article coverage is 0.569 for CIB, 0.387 for TMG, and 0.305 for OIH. These values explain part of the instability seen in the ticker-level results. The weakest panels are not simply weaker models. They are weaker information histories.

5.7 Auxiliary Experiments and Research Journey

The final sentiment-only benchmark was not the first experimental form of the project. The research process began with the industrial question of whether a sentiment signal could improve an existing EGX direction-forecasting model based on historical prices and technical features. That historical integration experiment used the Robin/Beltone internship codebase as an external baseline and compared the original historical features against the same feature set augmented with one-day-lagged bilingual sentiment features. The experiment was run for CIB, TMG, and OIH using logistic regression, XGBoost, and an MLP under a shared chronological test period.

Table 5.4: Summary of auxiliary experiments added to document the research journey.

Experiment	Purpose	Main outcome
Historical model integration	Test sentiment as an added feature in the Robin/Beltone EGX historical forecasting model	Sentiment improved raw accuracy in 5 of 9 ticker-model comparisons, but gains were conditional and not uniform
Sentiment-return association	Check whether lagged daily sentiment correlates directly with stock returns	No main lagged weighted-sentiment feature met the pre-specified Spearman and permutation threshold
Leakage and overfitting audit	Verify that final results are not inflated by future information or optimistic model selection	Leakage status passed across 45 panel checks and 43,880 validation/test rows; 15 of 45 rows were flagged for overfitting
AraBERT financial adaptation	Test whether Arabic financial news improves AraBERT masked-language modeling	Held-out MLM loss decreased from 0.9624 to 0.8468 relative to base AraBERT
Social-media feasibility	Determine whether social data can be added through free, ethical, official access	Reddit requires approved official access; X/Twitter broad public search is not free under the official API model

The historical integration result was mixed. Sentiment improved raw test accuracy in 5 of 9 ticker-model comparisons. The largest gain appeared for OIH with the MLP, where accuracy increased by 0.143, balanced accuracy by 0.106, and F1 by 0.309. The largest loss appeared for CIB with the MLP, where accuracy decreased by 0.105. Logistic regression was mostly flat or weaker after sentiment was added, while XGBoost showed small gains for CIB and OIH and a small loss for TMG. This result was useful as motivation, but it could not cleanly isolate the value of sentiment because the baseline already contained many price-derived predictors. That limitation is the reason the thesis moved toward the stricter sentiment-only benchmark reported above.

A second auxiliary experiment inspected the raw relationship between daily news sentiment and stock returns. Daily bilingual sentiment was shifted by one calendar day before alignment to trading dates, then correlated with same-day, 1-day, 3-day, and 5-day

returns. Pearson and Spearman correlations were computed, and p-values were estimated by block permutation with five-day blocks. No ticker-horizon pair for the main lagged weighted-sentiment feature met the pre-specified threshold of absolute Spearman correlation at least 0.10 and permutation p-value at most 0.05. The strongest displayed associations were still weak: OIH 3-day Spearman correlation was -0.127 with p-value 0.155, OIH 5-day was -0.109 with p-value 0.186, and TMG same-day was -0.102 with p-value 0.120. This supports a conservative interpretation: sentiment shows weak descriptive associations, but not a stable monotonic return relation by itself.

The Arabic sentiment component was also tested from a domain-adaptation perspective. Following the logic of FinBERT, a valid Arabic financial analogue would require finance-domain language-model adaptation followed by supervised sentiment fine-tuning on independent human-labeled Arabic financial examples. The available thesis corpus supports only the first stage. It contains 1,327 Arabic financial articles, split chronologically into 1,061 masked-language-model training articles and 266 validation articles. A short CPU-based AraBERT adaptation run completed for 20 steps, with training loss 0.9676 and validation loss 0.8218. A follow-up fixed-mask evaluation compared the adapted checkpoint with the original AraBERT model on the same corpus, as shown in Table 5.5. This is therefore reported as an MLM feasibility and domain-adaptation result, not as a completed Arabic FinBERT equivalent. Supervised Arabic financial sentiment fine-tuning remains future work because the present labels are model-generated outputs, not independent human annotations. [10, 11]

Table 5.5: Masked-language-model evaluation of base AraBERT and the financial-domain adapted checkpoint.

Split	Model	Examples	Masked tokens	Loss	Perplexity
Train	Base AraBERT	1061	40372	0.9413	2.563
Train	Financial-adapted AraBERT	1061	40372	0.8202	2.271
Validation	Base AraBERT	266	10292	0.9624	2.618
Validation	Financial-adapted AraBERT	266	10292	0.8468	2.332

Validation masked-language-model loss decreased from 0.9624 to 0.8468, while perplexity decreased from 2.618 to 2.332, corresponding to a 12.01% relative loss reduction. This supports the narrower claim that the adapted checkpoint models Arabic financial text better than the base checkpoint under the MLM objective. It does not prove better sentiment classification or better stock forecasting.

The final auxiliary investigation considered social-media sources. The requirement was to find a free, ethical, and officially approved route for EGX social-media sentiment. The conclusion was restrictive. Reddit can be considered only through official API or approved research access and with limits on redistribution and model training. X/Twitter was excluded from the current implementation path because broad public-read access is paid under the official API model, while unofficial scraping would be inappropriate for an academic thesis. Prior work on Arabic tweets and Arab stock-market indices, including EGX30, supports the relevance of this direction, but it does not remove the current access constraint. [30]

5.8 Interpretation

The final evidence supports a limited rather than optimistic conclusion. Bilingual news sentiment does contain some directional information, but the signal is not stable enough to serve as a strong standalone EGX forecasting system in the present dataset. The strongest accepted result is only marginally above random under balanced accuracy. Several apparently strong F1 values are driven by class skew rather than by robust up-versus-down discrimination.

This negative result is still informative. First, it justifies the conservative methodology because looser reporting could easily have selected the overfit OIH event-weighted model as the headline result. Second, it clarifies where later work should focus: richer relevance modeling, denser news coverage, Arabic financial-domain adaptation with human sentiment labels, official social-media access, and eventual integration with price-derived features rather than sentiment-only forecasting.

5.9 Chapter Summary

This chapter reported the concrete implementation settings and the final empirical results. The accepted benchmark uses bilingual Mubasher news, simple leakage-safe walk-forward models, and explicit validity checks. The strongest accepted result is OIH at the 5-day horizon under the all-news bilingual setting, but its balanced accuracy remains only slightly above random. Event filtering produces isolated gains without a consistent aggregate improvement. The auxiliary experiments complete the research journey: sentiment can conditionally help historical models, raw sentiment-return correlation remains weak, Arabic financial-domain adaptation improves MLM loss, and social media should remain future work under official access. The overall conclusion is that bilingual financial-news sentiment is informative but weak as a standalone forecasting signal in the current EGX dataset.

Chapter 6

Conclusion & Future Work

6.1 Conclusion

This thesis tested whether bilingual financial-news sentiment can support short-horizon EGX stock forecasting under a conservative and reproducible evaluation design. The study began from an industrial motivation: sentiment appeared useful as an additional signal inside an existing historical price-based forecasting model. Re-running that integration experiment showed conditional gains, but not uniform improvement. Sentiment improved raw accuracy in 5 of 9 ticker-model comparisons, with the strongest gain for OIH using an MLP, but it also reduced performance in other settings. This made the result useful as motivation rather than as a clean final claim.

The core contribution of the thesis is therefore the sentiment-only benchmark. The implemented pipeline collects Arabic and English Mubasher company news, routes English articles to FinBERT and Arabic articles to an AraBERT-based sentiment route, aggregates article-level sentiment into lagged daily features, and evaluates all-news, event-only, and event-weighted variants under chronological walk-forward validation. The final accepted result is modest. The strongest validity-passing candidate is OIH at the 5-day horizon under the all-news bilingual setting, with test accuracy 0.532, balanced accuracy 0.503, F1 score 0.630, and return sign accuracy 0.585. These values indicate weak informational value, not a strong standalone forecasting system.

The negative and mixed results are still scientifically useful. Leakage checks passed across 45 panel evaluations and 43,880 validation/test prediction rows, while overfitting analysis flagged 15 of 45 rows. The strongest raw-accuracy event-weighted candidate was therefore treated as exploratory instead of being promoted as the headline result. A separate sentiment-return correlation study also found no stable monotonic association under the pre-specified threshold. Together, these checks support the main conclusion: bilingual financial-news sentiment carries some signal in this EGX setting, but the signal is sparse, unstable, and highly sensitive to coverage, horizon, and validation discipline.

6.2 Future Work

The most immediate future step is to improve data coverage. The final audit found that TMG and OIH matched the currently available Mubasher listing pages, while CIB had one

additional Arabic listing page available. Re-running the resumable scraper before final submission would reduce this residual incompleteness. Broader source coverage should also be considered, but only if ticker linkage and timestamp quality remain auditable. Possible additions include official company disclosures, Egyptian Exchange announcements, and other financial-news providers with stable archives.

A second direction is Arabic financial-domain adaptation. The thesis prepared and tested the first stage of this process by running a short masked-language-model adaptation of AraBERT on 1,327 Arabic financial articles. The adapted checkpoint reduced held-out masked-language-model loss relative to base AraBERT, but a valid Arabic FinBERT-style model still requires independent human-labeled Arabic financial sentiment examples for supervised fine-tuning and evaluation. The next research step should therefore construct a small expert-labeled Arabic financial sentiment set with positive, negative, and neutral labels, evaluate inter-annotator agreement, fine-tune the adapted encoder with a classification head, and compare it against the current AraBERT route on held-out human labels. Using the current model-generated labels as ground truth would only distill the existing classifier and would not prove improved sentiment quality.

A third direction is controlled integration with market features. The historical integration experiment suggests that sentiment may help selected model-ticker pairs when added to price-derived features, but that experiment should be repeated with a fully unified target definition, walk-forward protocol, and ablation design before making stronger claims. A stronger follow-up would compare price-only, sentiment-only, and price-plus-sentiment models under the same target, embargo, validation, threshold-selection, and test-period rules.

Fourth, the event-aware layer can be expanded beyond keyword-style categories. The present event labels are useful as a lightweight relevance signal, but they do not capture event materiality, novelty, or whether the article describes expected or surprising information. A future system could train a supervised event classifier, estimate event severity, and separate company-specific events from market-wide coverage.

Finally, social media remains future work rather than a completed experiment. Reddit is potentially feasible only through official API or approved research access with strict data-handling limits. X/Twitter is not a free broad-search option under the current official API model. If approved access becomes available, the next design should collect posts only through official channels, avoid redistribution of raw user content, map Arabic and English aliases for EGX tickers, and compare news-only, social-only, news-plus-social, and ensemble models under the same leakage-safe evaluation framework used in this thesis.

Appendix

Appendix A

Lists

List of Figures

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